

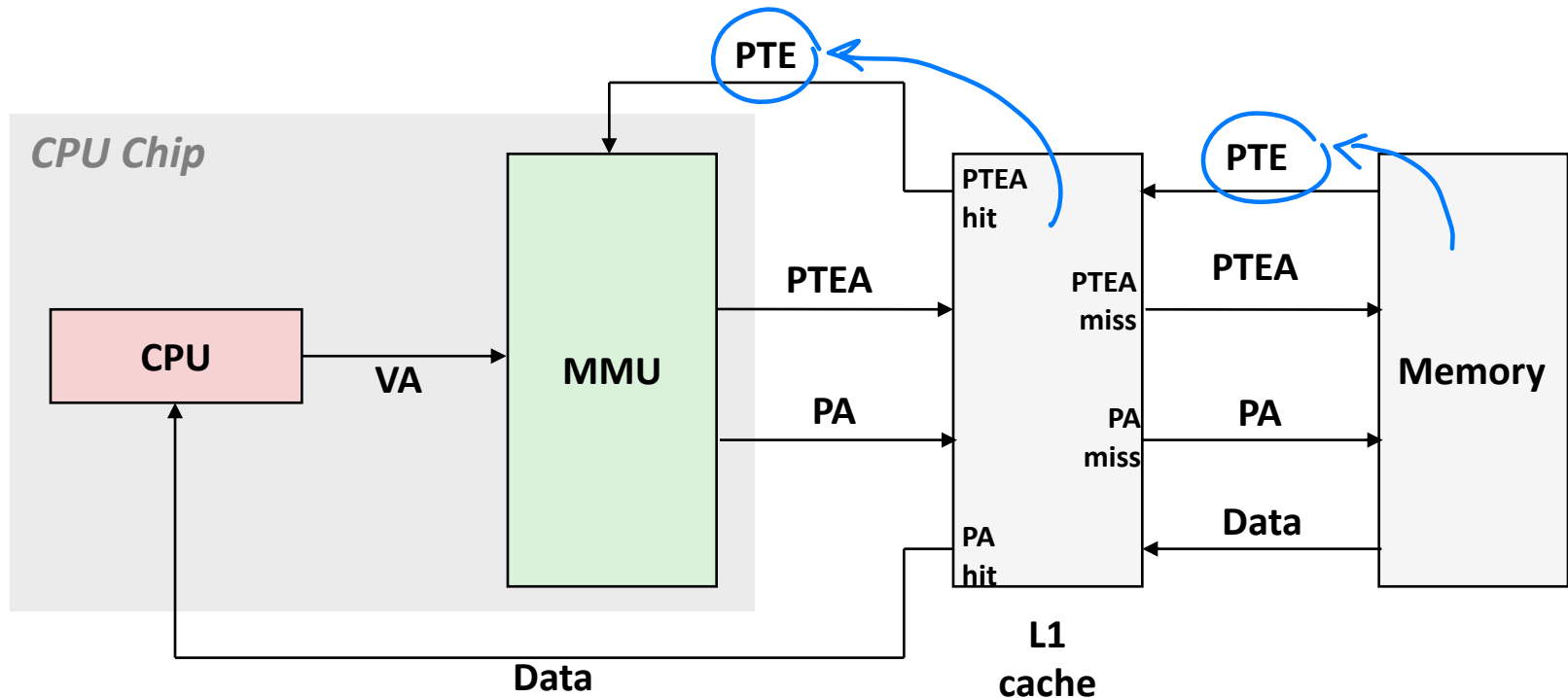
Question #1

- Are the PTEs cached like other memory accesses?

✓. 页表常驻内存. 缓存在 SRAM 缓存上.

- Yes (and no: see next question)

Page tables in memory, like other data



VA: virtual address, PA: physical address, PTE: page table entry, PTEA = PTE address

Question #2

■ Isn't it slow to have to go to memory twice every time?

■ Yes, it would be... so, real MMUs don't

✓, 所以MMU内部有一个储存最近使用的PTE的小缓存: TLB.

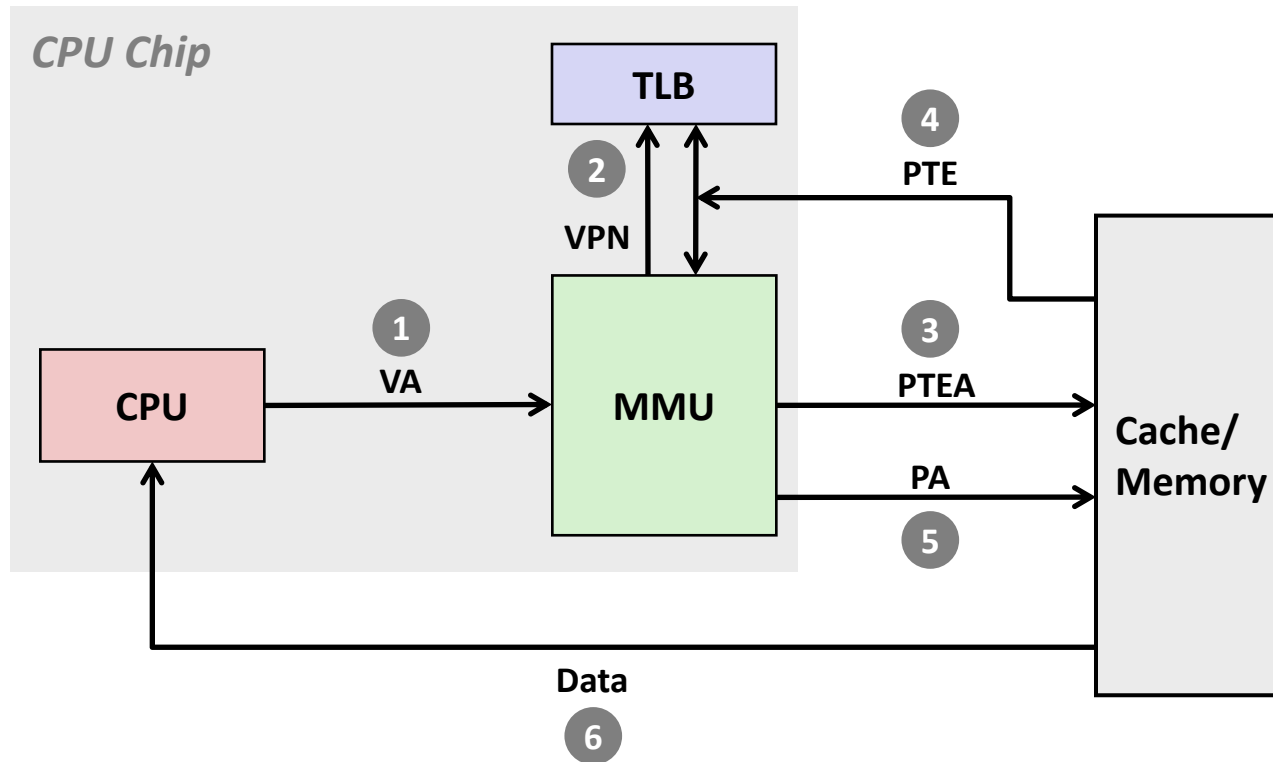
"Lookaside Buffer"

TLB实际缓存了最近使用的虚拟页号到物理页号的映射.

Speeding up Translation with a TLB

- Page table entries (PTEs) are cached in L1 like any other memory word
 - PTEs may be evicted by other data references
 - PTE hit still requires a small L1 delay
- Solution: *Translation Lookaside Buffer* (TLB)
 - Small, dedicated, super-fast hardware cache of PTEs in MMU
 - Contains complete page table entries for small number of pages

TLB Miss



A TLB miss incurs an additional memory access (the PTE)

Fortunately, TLB misses are rare. Why?

① Locality

② TLB 相联度高.
几乎没有 conflict miss

Question #3

- Isn't the page table huge? How can it be stored in RAM?

Hierarchy. 采用多级页表

- Yes, it would be... so, real page tables aren't simple arrays

* Huge: 假设 page size = 4KB = 2^{12} bytes

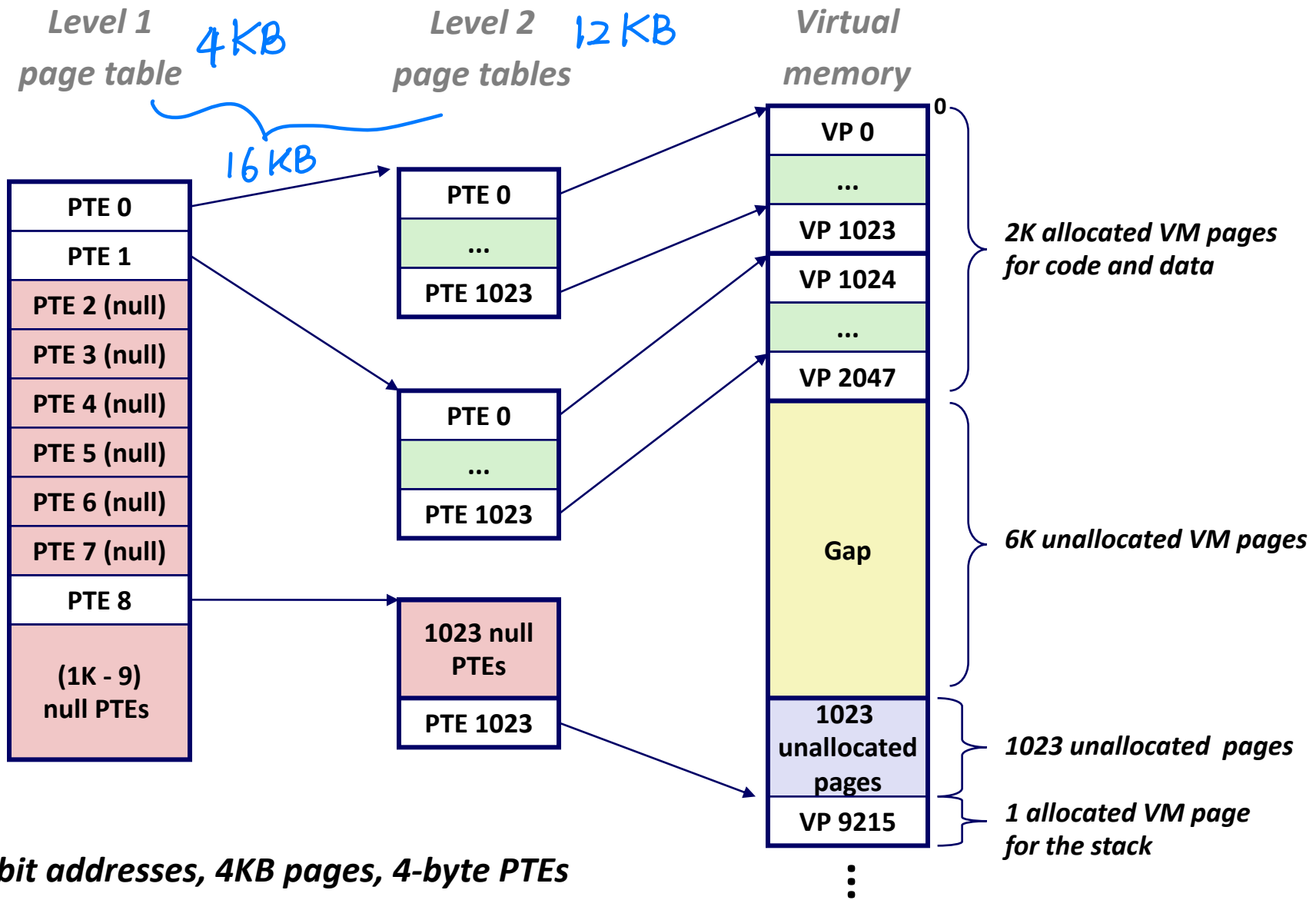
64位: 地址空间 $N = 2^{64}$ bytes

$$\Rightarrow \text{PTE 数} = \frac{2^{64}}{2^{12}} = 2^{52}$$

一个PTE 8字节

$$\Rightarrow \text{页表大小} \cdot 2^{52} \times 8 = 2^{55} \text{ bytes} ?? \quad (1\text{TB} = 2^{40} \text{ bytes})$$

A Two-Level Page Table Hierarchy



Question #4

- Aren't the TLB contents wrong after a context switch?

✓. 每个进程有一张独立页表.

多个进程可能用相同虚拟地址, 映射到不同物理地址

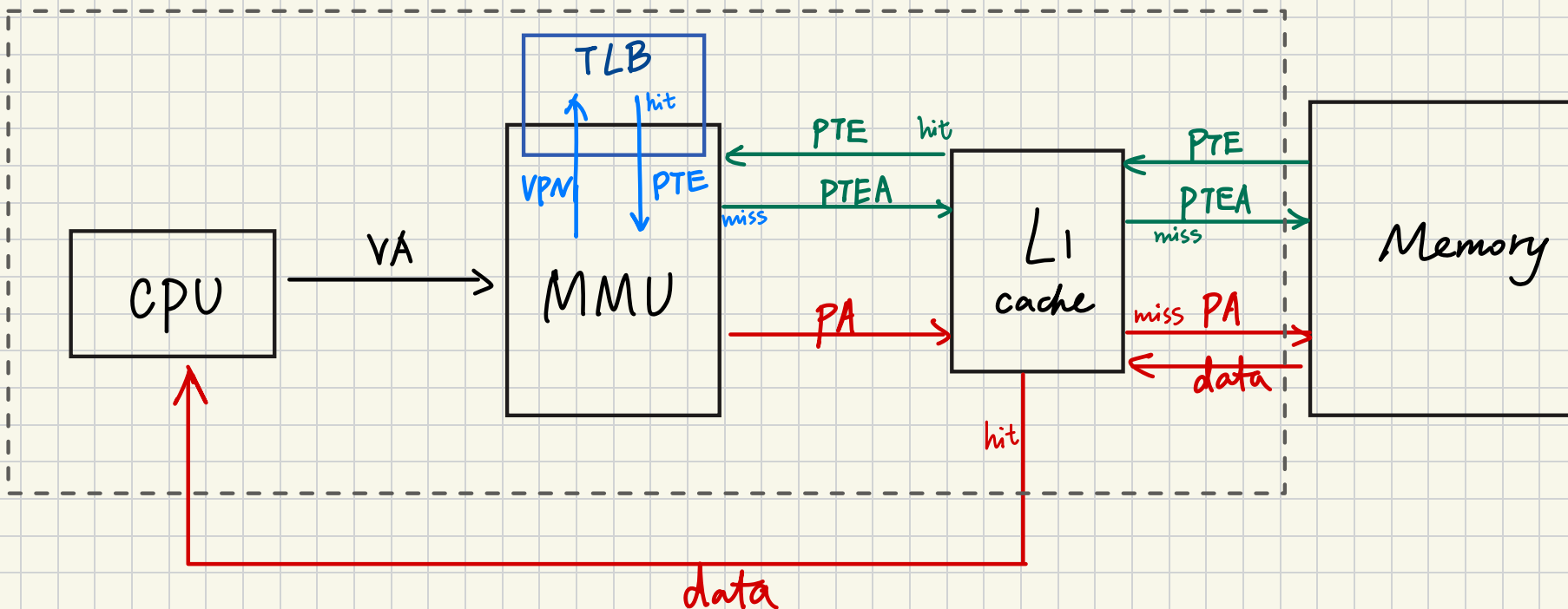
- Yes, they would be, so something must be done.. (e.g. $0x400000$: Unix 程序加载地址)
 - Option 1: flush TLB on context switch
 - Option 2: associate a process ID with each TLB entry

① 每次 Context Switch 把 TLB 中每个 PTE 的 valid bit 置 0, 热身 访问页表

② PTE 里增加几位储存 PID / ASID: Address Space Identifier.

Memory 访存

CPU chip



* TLB miss 后, PTEA 哪来的?

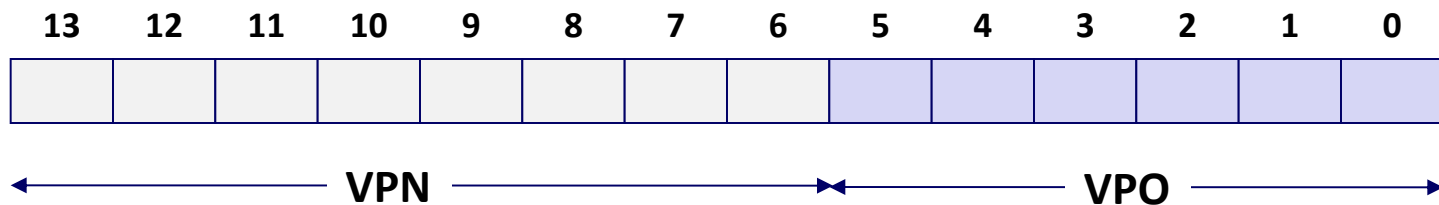
页表基址 (OS 写入专用寄存器内) + $VPN * \text{sizeof}(PTE)$.

物理地址.

Simple Memory System Example

■ Addressing

- 14-bit virtual addresses $N = 2^{14}$
- 12-bit physical address $M = 2^{12}$
- Page size = $\frac{64 \text{ bytes}}{2^6}$ $\Rightarrow$ PTE 数 = $\frac{2^{14}}{2^6} = 2^8 = 256$.

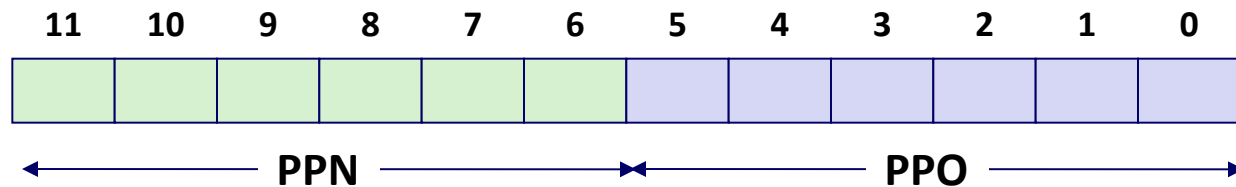


Virtual Page Number

$\sim$ PTE, 8位

Virtual Page Offset

$\sim$ Page size, 6位

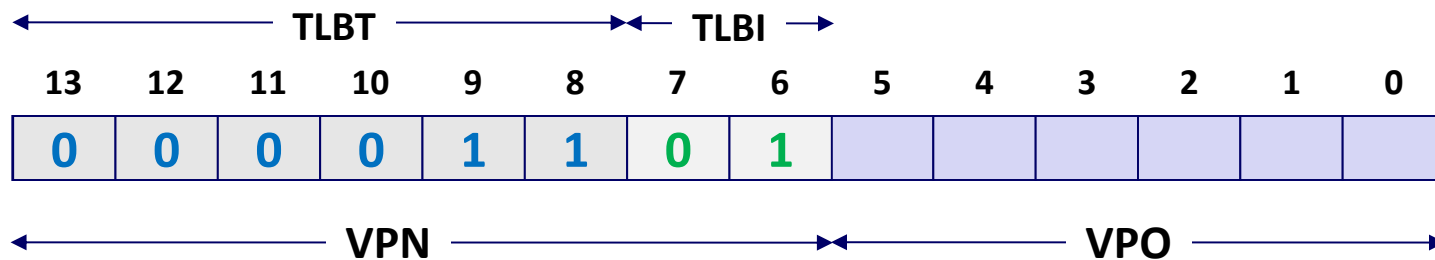


Physical Page Number

Physical Page Offset

Simple Memory System TLB

- 16 entries → 容量
- 4-way associative → 4路相联. $L=4$



$$\text{VPN} = 0b1101 = 0x0D$$

Translation Lookaside Buffer (TLB)

Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	–	0	09	0D	1	00	–	0	07	02	1
1	03	2D	1	02	–	0	04	–	0	0A	–	0
2	02	–	0	08	–	0	06	–	0	03	–	0
3	07	–	0	03	0D	1	0A	34	1	02	–	0

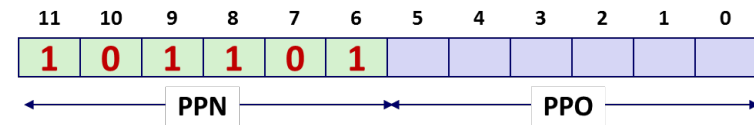
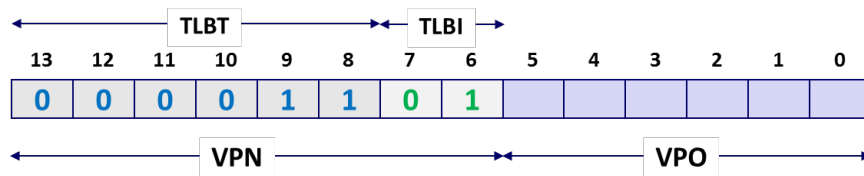
Simple Memory System Page Table

Only showing the first 16 entries (out of 256)

<i>VPN</i>	<i>PPN</i>	<i>Valid</i>
00	28	1
01	—	0
02	33	1
03	02	1
04	—	0
05	16	1
06	—	0
07	—	0

<i>VPN</i>	<i>PPN</i>	<i>Valid</i>
08	13	1
09	17	1
0A	09	1
0B	—	0
0C	—	0
0D	2D	1
0E	11	1
0F	0D	1

0x0D → 0x2D



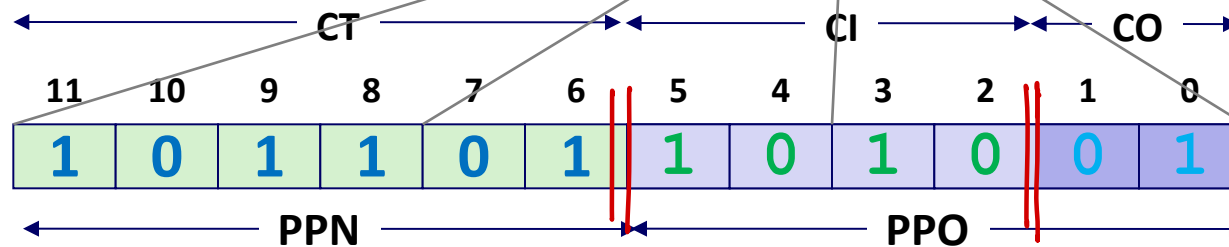
Simple Memory System Cache

■ 16 lines, 4-byte block size $S=4$ $b=2$

■ Physically addressed

■ Direct mapped: $L=1$

$V[0b00001101101001] = V[0x369]$
 $P[0b101101101001] = P[0xB69] = 0x15$

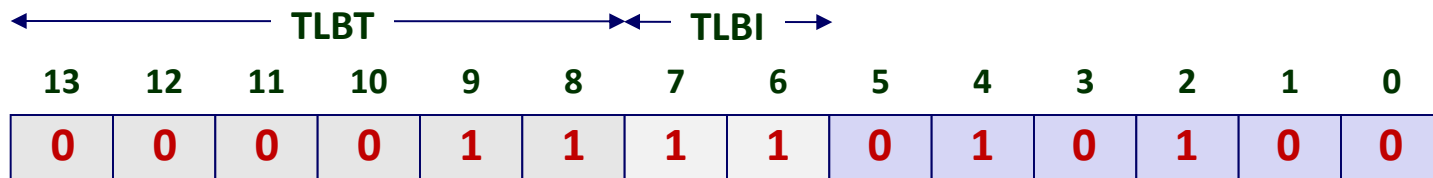


Idx	Tag	Valid	B0	B1	B2	B3
0	19	1	99	11	23	11
1	15	0	–	–	–	–
2	1B	1	00	02	04	08
3	36	0	–	–	–	–
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	–	–	–	–
7	16	1	11	C2	DF	03

Idx	Tag	Valid	B0	B1	B2	B3
8	24	1	3A	00	51	89
9	2D	0	–	–	–	–
A	2D	1	93	15	DA	3B
B	0B	0	–	–	–	–
C	12	0	–	–	–	–
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	–	–	–	–

Address Translation Example: TLB/Cache Hit

Virtual Address: 0x03D4

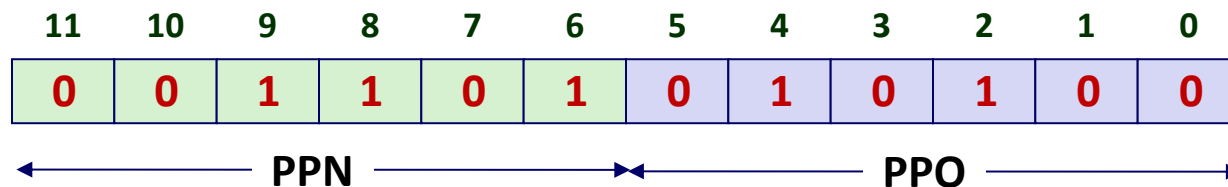


VPN 0x0F TLBI 0x3 TLBT 0x03 TLB Hit? Y Page Fault? N PPN: 0x0D

TLB

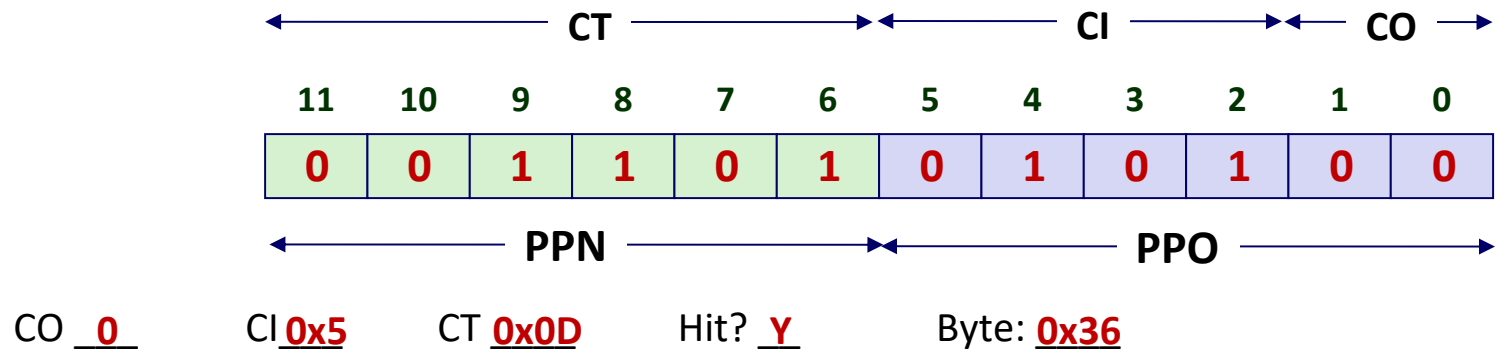
Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	–	0	09	0D	1	00	–	0	07	02	1
1	03	2D	1	02	–	0	04	–	0	0A	–	0
2	02	–	0	08	–	0	06	–	0	03	–	0
3	07	–	0	03	0D	1	0A	34	1	02	–	0

Physical Address



Address Translation Example: TLB/Cache Hit

Physical Address



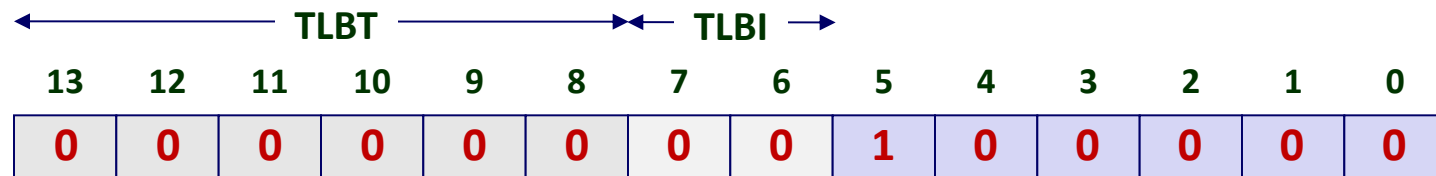
Cache

Idx	Tag	Valid	B0	B1	B2	B3
0	19	1	99	11	23	11
1	15	0	–	–	–	–
2	1B	1	00	02	04	08
3	36	0	–	–	–	–
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	–	–	–	–
7	16	1	11	C2	DF	03

Idx	Tag	Valid	B0	B1	B2	B3
8	24	1	3A	00	51	89
9	2D	0	–	–	–	–
A	2D	1	93	15	DA	3B
B	0B	0	–	–	–	–
C	12	0	–	–	–	–
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	–	–	–	–

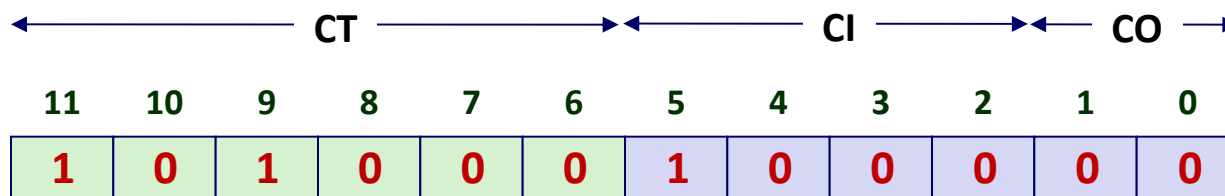
Address Translation Example: TLB/Cache Miss

Virtual Address: 0x0020



VPN 0x00 TLBI 0 TLBT 0x00 TLB Hit? N Page Fault? N PPN: 0x28

Physical Address



PPN PPO

CO 0 CI 0x8 CT 0x28 Hit? Byte:

Page table

VPN	PPN	Valid
00	28	1
01	–	0
02	33	1
03	02	1
04	–	0
05	16	1
06	–	0
07	–	0

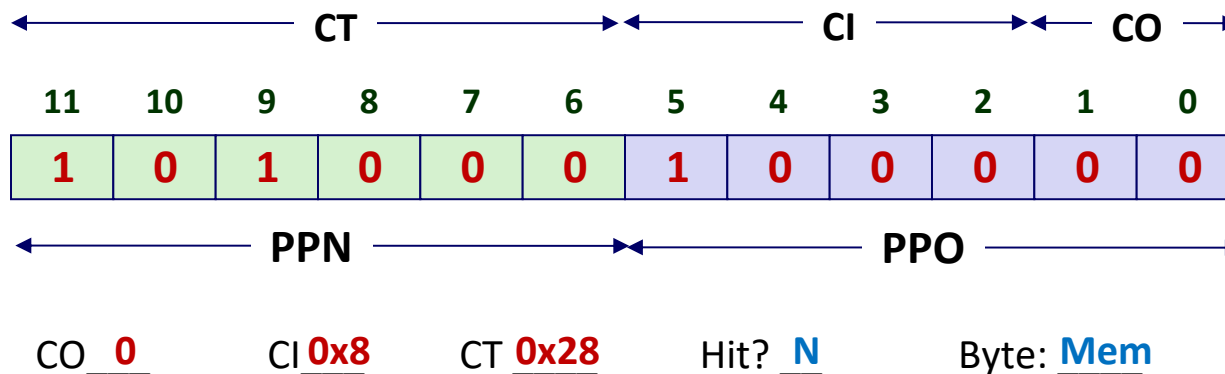
Address Translation Example: TLB/Cache Miss

Cache

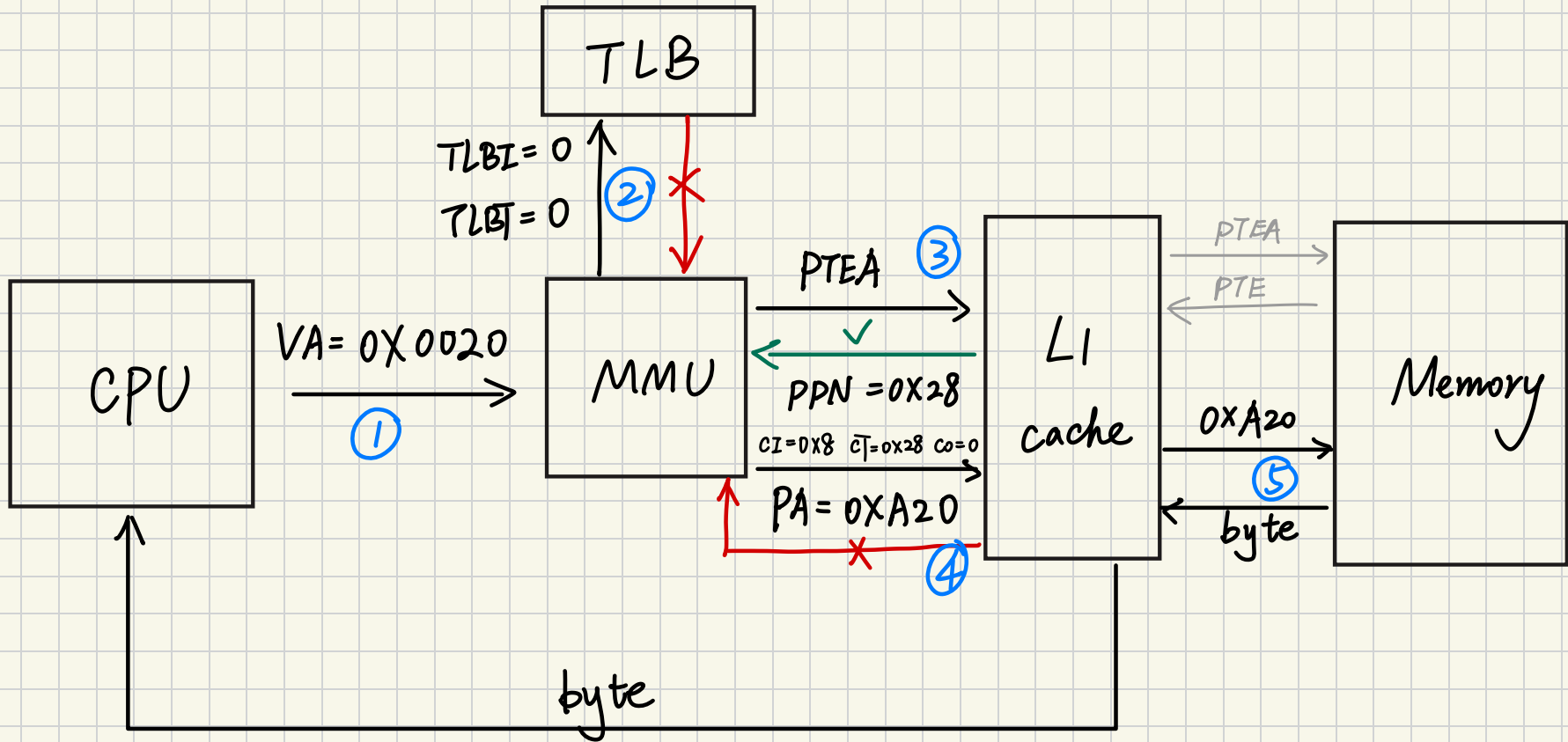
Idx	Tag	Valid	B0	B1	B2	B3
0	19	1	99	11	23	11
1	15	0	–	–	–	–
2	1B	1	00	02	04	08
3	36	0	–	–	–	–
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	–	–	–	–
7	16	1	11	C2	DF	03

Idx	Tag	Valid	B0	B1	B2	B3
8	24	1	3A	00	51	89
9	2D	0	–	–	–	–
A	2D	1	93	15	DA	3B
B	0B	0	–	–	–	–
C	12	0	–	–	–	–
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	–	–	–	–

Physical Address

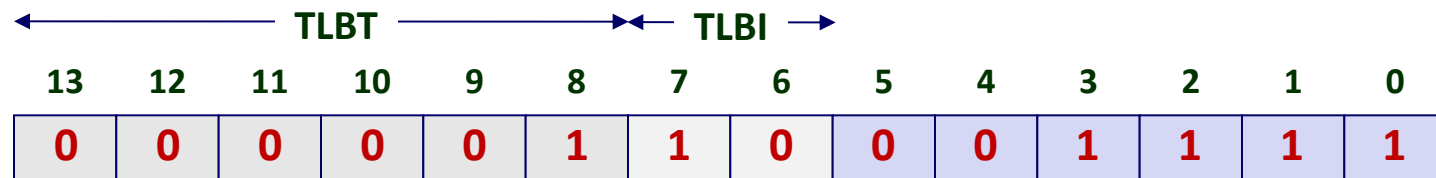


TLB & Cache Miss



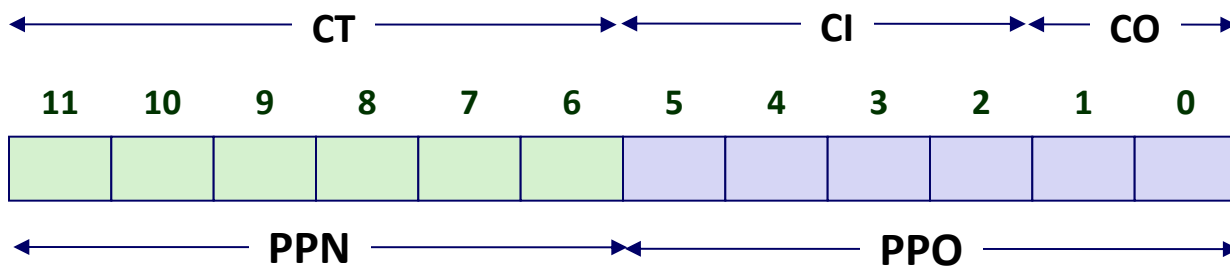
Address Translation Example: Page Fault

Virtual Address: 0x018F



VPN 0x06 TLBI 0x2 TLBT 0x01 TLB Hit? N Page Fault? Y PPN: TBD

Physical Address



CO ____ CI ____ CT ____ Hit? ____ Byte: ____

Page table

VPN	PPN	Valid
00	28	1
01	–	0
02	33	1
03	02	1
04	–	0
05	16	1
06	–	0
07	–	0

Problem 5. (10 points):

Assume a System that has

1. A two way set associative TLB $L=2$.
2. A TLB with 8 total entries $S=4$
3. 2^8 byte page size $VPO: 8 \text{ 位}$
4. 2^{16} bytes of virtual memory $PTE \text{ 数}: 2^8 = 256 \Rightarrow VPN \text{ 8 位}, VPO \text{ 8 位}$
5. one (or more) boats $??$

5. one (or more) boats
在计算机系统，虚拟内存，TLB 中都没有任何参数叫 "boat"。
这题是故意用这个词。
• "boat" 不会影响计算
• 是故意加的一条信息来混淆
• 目的就是让学生看到时笑一下，不要以为是考点

TLBT: 6 TLBI: 2 PPO: 8

TLB			
Index	Tag	Frame Number	Valid
0	0x13	0x30	1
	0x34	0x58	0
1	0x1F	0x80	0
	0x2A	0x72	1
2	0x1F	0x95	1
	0x20	0xAA	0
3	0x3F	0x20	1
	0x3E	0xFF	0

A. Use the TLB to fill in the table. Strike out anything that you don't have enough information to fill in.

Virtual Address	Physical Address
0x7E85	0x9585
0xD301	/
0x4C20	0x3020
0xD040	/
/	0x5830

(1)

VA: 0x7E85

$$\begin{array}{c} \text{VPN} \qquad \qquad \qquad \text{VPO} \\ \hline 01111100 \mid 0000101 \\ \hline \text{TLBT} \quad \text{TLBI} \qquad \qquad \text{PPO} \\ 0x1F \quad 0x2 \qquad \qquad 0x85 \end{array}$$

hit $\Rightarrow PPN = 0x95 \quad PPO = 0x85$

(2) VA: 0xD301

$$\begin{array}{c} \hline 10110011 \quad 00000000 \mid \\ \hline \text{TLBT} \quad \text{TLBI} \\ 0x2C \quad 0x3 \quad \text{TLB miss} \end{array}$$

(3) PA: 0x3020

$$\begin{array}{c} \text{PPN} \qquad \qquad \qquad \text{PPO} \\ \hline 00110000 \quad 00100000 \\ \hline \downarrow \\ \text{TLBT} = 0x13 \quad \text{TLBI} = 0 \quad \text{VPO} \\ \Rightarrow \text{VPN} \cdot 01001100 = 0x4C \end{array}$$

(4) VM: 0xD040

$$\begin{array}{c} \hline 11010000 \quad \dots \\ \hline \text{TLBT} \quad \text{TLBI} \\ 0x34 \quad 0 \quad \text{Valid} = 0. \text{ miss} \end{array}$$

(5) PA: 0x5830

$$\begin{array}{c} \hline 01011000 \quad 00110000 \\ \hline \text{PPN} = 0x58. \text{ Valid} = 0. \text{ X} \end{array}$$